

Xuelin Hu^a, Xiaoqin Fu^a, Youjing Fu^b, Jingchao Wang^c, Ruishen Liu^d, Charles Jumaa Katila^e and Pengming Hu^{c,*}

^a*Liuzhou Railway Vocational Technical College, No. 2 Wenyuan Road, Yufeng District, Liuzhou, 545000, Guangxi Zhuang Autonomous Region, China*

^b*College of Earth and Environmental Sciences, Lanzhou University, Lanzhou, 730000, China*

^c*School of Computer Science, Zhongyuan University of Technology, Zhengzhou, 450007, China*

^d*School of Artificial Intelligence, Luoyang Normal University, Luoyang, 471000, China*

^e*School of Computing and Mathematics, The Co-operative University of Kenya, Kenya*

ABSTRACT

ORCID identifiers

Xuelin Hu: 0000-0002-4475-3034.

Xiaoqin Fu: 0009-0003-5123-8393.

Youjing Fu: 0009-0004-1369-7879.

Funding

This work was supported by the Science and Technology Research Project of Henan Province (grant no. 262102211055). The funder had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript. This work was also funded by the Key Scientific Research Project of Colleges and Universities in Henan Province (grant no. 27AQ520018).

*Corresponding author



huxuelinai@gmail.com (X. Hu); huxl@ltzy.edu.cn (X. Hu);

xiaoqin.fu@qq.com (X. Fu); fuxiaoqin@ltzy.edu.cn (X. Fu);

fuyj2025@lzu.edu.cn (Y. Fu); static@zut.edu.cn (J. Wang);

liuruishen@lynu.edu.cn (R. Liu); ckatila@cuk.ac.ke (C.J. Katila);

hupengming@zut.edu.cn (P. Hu)

ORCID(s):